

Double DES (2DES)

1. Aim

To implement the Double DES (2DES) encryption technique using two independent keys applied successively, and to study how encrypting data twice with two different keys strengthens a symmetric cipher.

2. Algorithm

1. Start the program.
2. Declare variables for plaintext, key1, key2 and cipher.
3. Read the plaintext value entered by the user.
4. Read the first key (key1) entered by the user.
5. Read the second key (key2) entered by the user.
6. Encrypt the plaintext by performing XOR with key1, then XOR the result with key2 to obtain the final cipher.
7. Display the encrypted value.
8. Decrypt the cipher by performing XOR with key2 followed by XOR with key1, in the reverse order of encryption.
9. Display the decrypted value and stop the program.

3. Program

```
#include <stdio.h>

int main() {
    int plaintext, key1, key2, cipher;

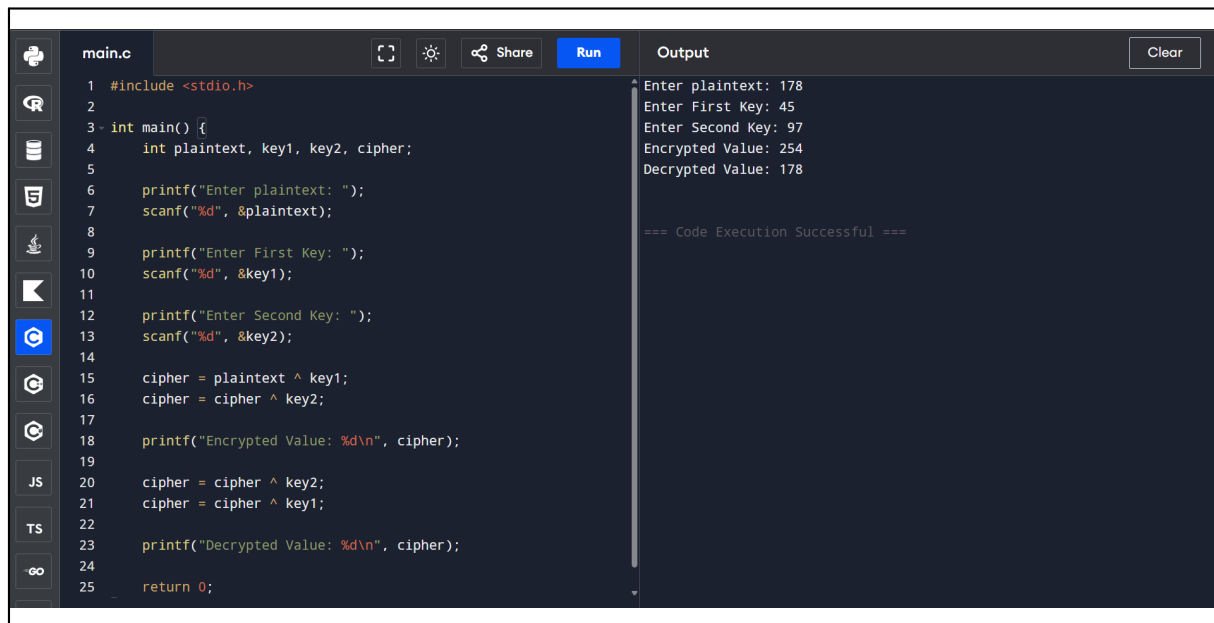
    printf("Enter plaintext: ");
    scanf("%d", &plaintext);
    printf("Enter First Key: ");
    scanf("%d", &key1);
    printf("Enter Second Key: ");
    scanf("%d", &key2);

    cipher = plaintext ^ key1;
    cipher = cipher ^ key2;
    printf("Encrypted Value: %d\n", cipher);

    cipher = cipher ^ key2;
    cipher = cipher ^ key1;
    printf("Decrypted Value: %d\n", cipher);

    return 0;
}
```

4. Output Screenshot



The screenshot shows a code editor with a file named `main.c`. The code implements a Double DES (2DES) encryption and decryption process. It prompts the user for a plaintext value and two keys, then calculates the encrypted value and finally the decrypted value. The output window on the right shows the results of the execution with the input values 178, 45, and 97, resulting in an encrypted value of 254 and a decrypted value of 178. A message at the bottom of the output window states "=== Code Execution Successful ===".

```
1 #include <stdio.h>
2
3 int main() {
4     int plaintext, key1, key2, cipher;
5
6     printf("Enter plaintext: ");
7     scanf("%d", &plaintext);
8
9     printf("Enter First Key: ");
10    scanf("%d", &key1);
11
12    printf("Enter Second Key: ");
13    scanf("%d", &key2);
14
15    cipher = plaintext ^ key1;
16    cipher = cipher ^ key2;
17
18    printf("Encrypted Value: %d\n", cipher);
19
20    cipher = cipher ^ key2;
21    cipher = cipher ^ key1;
22
23    printf("Decrypted Value: %d\n", cipher);
24
25    return 0;
}
```

Output:

```
Enter plaintext: 178
Enter First Key: 45
Enter Second Key: 97
Encrypted Value: 254
Decrypted Value: 178

=== Code Execution Successful ===
```

5. Sample Input / Output

```
Enter plaintext: 100
Enter First Key: 23
Enter Second Key: 45
Encrypted Value: 94
Decrypted Value: 100
```

6. Result

The Double DES (2DES) program applying two successive XOR key operations was implemented and executed successfully, and the original plaintext was correctly recovered after decryption.